

Manikanta Reddy Venna

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EDUCATION

University of South Florida, B.S. in Computer Science | GPA: 4.0/4.0

Expected Graduation: May 2027

SKILLS

Languages: Python, C#, TypeScript, JavaScript, C/C++, SQL

Backend, Frontend & Data: FastAPI, React, Node.js/Express, .NET Core, REST APIs, PostgreSQL, MongoDB, JWT

AI/ML: OpenAI API, RAG pipelines, ChromaDB, embeddings, LangChain, TensorFlow.js, PyTorch

Tools & Cloud: Docker, AWS (S3, EC2), GitHub Actions, CI/CD, Git, Linux/Unix

EXPERIENCE

Software Engineering Intern (Backend) | Automox

June 2025 – August 2025 (Remote)

- Implemented backend REST API endpoints supporting Automox's July 2025 Device Inventory rollout, automating device policy deployment and patch-remediation workflows — reducing manual configuration time by 40% across managed device operations.
- Built a Python diagnostics utility adopted by backend engineers to automate log and system-state collection across 5+ components, cutting average production incident debugging time from 2 hours to 20 minutes
- Resolved 8+ OS-specific production edge cases during the Agent 1.x to 2.x platform migration by tracing API request flows, diagnosing PostgreSQL query bottlenecks, and validating fixes across cross-platform managed device environments.
- Shipped backend changes through CI/CD pipelines with 95%+ test coverage, maintaining regression-tested reliability across device inventory, policy enforcement, and patch automation workflows throughout the migration cycle

C Programming Teaching Assistant (Unix/Linux) | University of South Florida

August 2025 – Present

- Debugged segmentation faults, memory leaks, and pointer errors across 100+ C codebases in Unix/Linux; performed structured code reviews that improved average submission scores by 15% through targeted, actionable feedback.
- Reviewed and validated submissions using Gradescope (autograding platform); identified correctness bugs, edge-case failures, and coding-standard violations within 2 weeks.
- Coordinated a 4-person TA team to standardize grading criteria and evaluation benchmarks across 150+ student submissions; designed regrade review workflow adopted as the course-wide standard for code quality assessment.

PROJECTS

EchoSoul — Persona Reasoning Engine | Python, FastAPI, React, TypeScript, PostgreSQL, ChromaDB, OpenAI API, ElevenLabs, Docker

- Built an app where users hold real conversations with AI-modeled personas — the system builds a psychological model from source material capturing how each persona thinks, reacts, and decides across any situation.
- Engineered a RAG pipeline using OpenAI embeddings, ChromaDB metadata-filtered retrieval, and PostgreSQL profile storage — ingesting 500+ source passages across 2 personas to ground every response in persona-specific traits and evidence instead of generic LLM output.
- Achieved 87% average persona-consistency across 50 benchmark prompts using an automated evaluator for emotion alignment, evidence grounding, and response consistency; integrated OpenAI STT and ElevenLabs TTS for voice-based interaction.

Smart Home Inventory & Expiration Tracker | C#, .NET Core, PostgreSQL, Entity Framework Core, Hangfire

- Built a barcode-driven home inventory system that prevents food waste through expiration alerts, smart shopping lists, GS1 barcode parsing with fallback validation, and consumption-rate forecasting.
- Developed a .NET Core REST API with PostgreSQL and Entity Framework Core; designed normalized schemas, usage logs, transactions, and indexes; validated core backend logic with xUnit tests reaching 75%+ coverage.

NeuralCloud — Browser ML Training & Deployment Platform | React, TensorFlow.js, WebGPU, Node.js, PostgreSQL, AWS S3

- Built a browser-based ML training platform that runs without a separate ML environment, supporting CSV preprocessing, real-time training metrics, and reproducible model artifact export across WebGPU, WebGL, and CPU backends.
- Implemented TensorFlow.js backend selection and benchmarking, achieving 3.2x faster inference with WebGPU vs. CPU baseline; engineered a deployment API with PostgreSQL model versioning, AWS S3 artifact storage, Dockerized services, and automated deploy/predict tests.

Leadership & Recognition

Student Government Senator — University of South Florida

August 2025 – Present

- Represent 50,000+ students in Student Government; recognized as Senator of the Month for leading committee initiatives and cross-campus senate meeting coordination.